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Dop: 21/08/2026

Experiment: Measurement of Signal Strength vs. Distance for a Wi-Fi Router

1. Aim

To **measure the Wi-Fi signal strength (RSSI) at different distances from a Wi-Fi router** and study the relationship between signal strength and distance.

2. Objectives

1. To understand the concept of Wi-Fi signal strength.
2. To measure RSSI at different distances from a Wi-Fi router.
3. To observe the variation of signal strength with distance.
4. To plot a graph of **distance vs. RSSI**.
5. To determine how distance and obstacles affect Wi-Fi communication quality.

3. Apparatus / Requirements

- Wi-Fi Router
- Laptop / Desktop / Smartphone
- Wi-Fi Analyzer application or software
- Measuring tape
- Open space or laboratory
- Graph paper / Excel / MATLAB (optional)

4. Theory

Wi-Fi is a wireless communication technology that uses radio waves to transmit data between a wireless router and connected devices. The strength of the received Wi-Fi signal depends on several factors such as **distance, frequency, obstacles, interference, antenna characteristics, and transmission power**.

When a device moves away from the Wi-Fi router, the received signal generally becomes weaker. The signal strength is commonly represented using **RSSI (Received Signal Strength Indicator)** and is expressed in **dBm**.

RSSI values are normally negative. A value closer to zero represents a stronger signal.

RSSI	Signal Quality
-30 dBm	Excellent
-50 dBm	Very good
-60 dBm	Good
-70 dBm	Fair
-80 dBm	Weak
-90 dBm	Very weak

As the distance between the router and receiving device increases, the received power decreases. Under ideal free-space conditions, this can be represented by the **Free Space Path Loss** relationship:

$$P_r \propto d^{21}$$

where:

- P_r = received signal power
- d = distance between transmitter and receiver

In practical indoor Wi-Fi environments, walls, furniture, people, reflections and interference also affect the measured RSSI. Therefore, the measured values may not decrease perfectly or smoothly with distance.

5. Experimental Setup

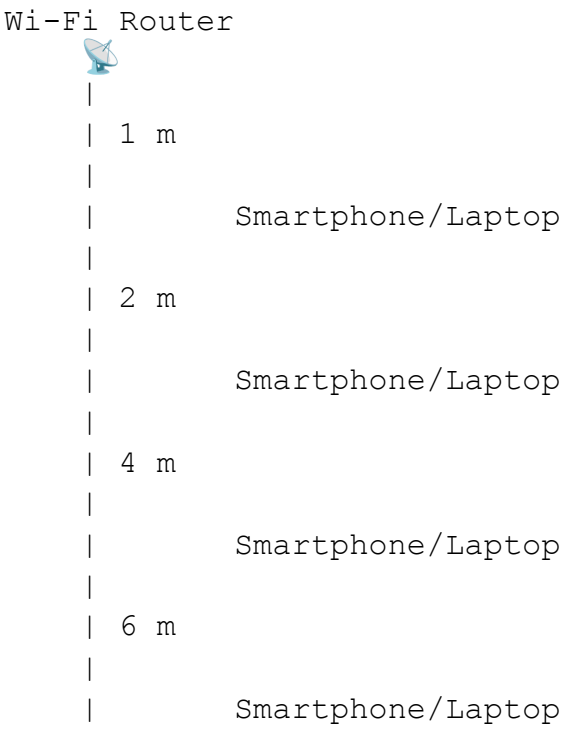
The Wi-Fi router is kept at a fixed position. The laptop or smartphone is placed at different distances from the router.

Example distances:

1 m → 2 m → 4 m → 6 m → 8 m → 10 m

At each distance, the Wi-Fi signal strength (RSSI) is recorded using a Wi-Fi analyzer.

Simple setup



Keep the **router, receiving device, Wi-Fi channel and orientation** as constant as possible during the experiment.

6. Procedure

1. Switch ON the Wi-Fi router.
2. Connect the smartphone/laptop to the selected Wi-Fi network.
3. Open a **Wi-Fi Analyzer** application/software.
4. Note the SSID of the selected Wi-Fi network.
5. Keep the router at a fixed location.
6. Measure a distance of **1 m** from the router.
7. Place the receiving device at this location.
8. Observe and record the RSSI value in dBm.
9. Repeat the measurement at **2 m, 4 m, 6 m, 8 m and 10 m**.
10. Record all readings in the observation table.
11. Calculate the average RSSI if multiple readings are taken at each distance.
12. Plot a graph of **Distance (m) vs. RSSI (dBm)**.
13. Analyze how the signal strength changes as the distance increases.

7. Observation Table

The following values are **sample readings**. Actual values will depend on the router and laboratory environment.

Sr. No.	Distance (m)	RSSI (dBm)	Signal Quality
1	1	-40	Excellent
2	2	-45	Very Good
3	4	-52	Very Good
4	6	-60	Good
5	8	-68	Good
6	10	-75	Fair
7	12	-81	Weak
8	14	-86	Very Weak

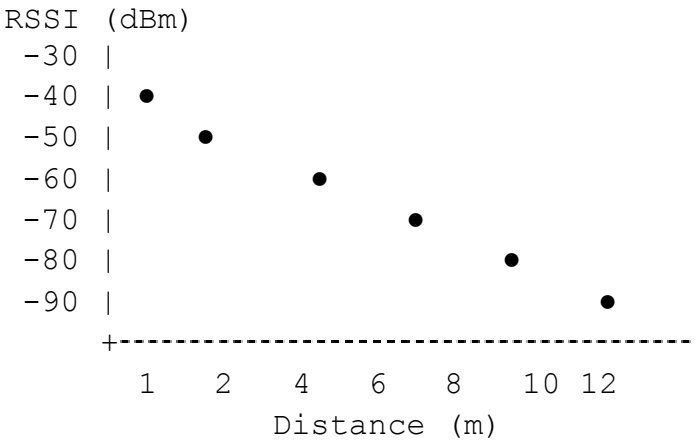
Important: RSSI becomes **more negative** as the signal becomes weaker.

8. Graph

Plot:

- **X-axis:** Distance in metres (m)
- **Y-axis:** RSSI in dBm

The graph generally shows that the RSSI decreases as the distance from the router increases.



The actual graph obtained in a laboratory may fluctuate because of walls, furniture, multipath propagation and other Wi-Fi networks.

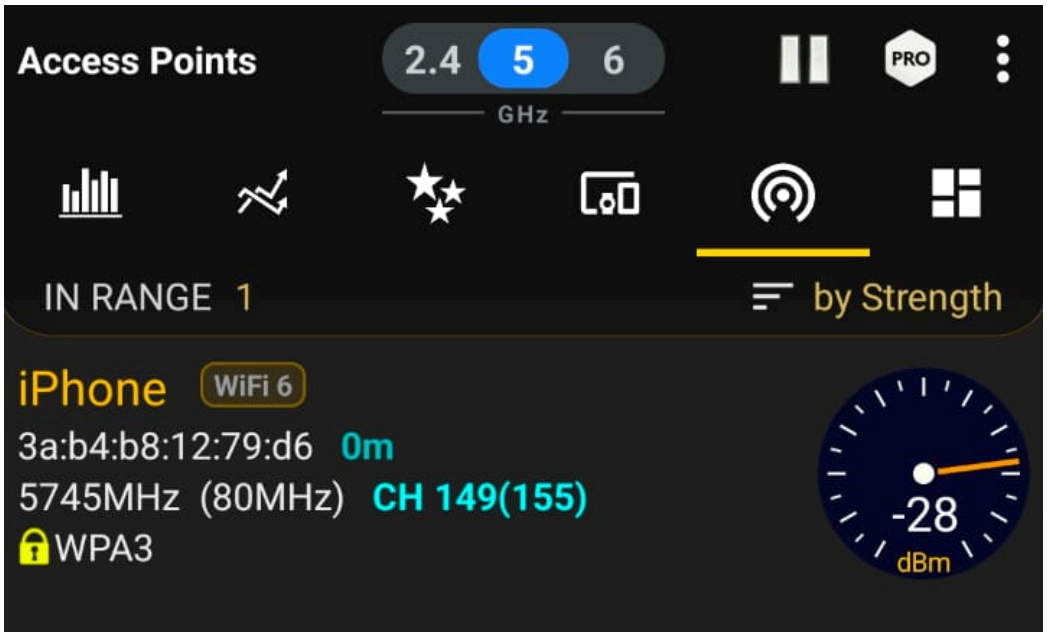
9. Result

The Wi-Fi signal strength was measured at different distances from the router. It was observed that **RSSI generally decreases as the distance between the Wi-Fi router and receiving device increases**. Hence, the Wi-Fi signal becomes weaker with increasing distance.

10. Conclusion

The experiment demonstrates the relationship between **Wi-Fi signal strength and distance**. The received signal is strongest near the router and gradually becomes weaker as the receiving device moves farther away. In practical environments, obstacles and interference can cause additional variations in signal strength.


Access Points
2.4
5
6
GHz
PRO
IN RANGE 4
by Strength
vivo V29e
WiFi 4
3a:1e:4a:6c:95:a7 8m
2467MHz (20MHz) CH 12
WPA2
-58
dBm
realme C53
WiFi 4
d6:4c:6e:ed:da:cd 12m
2442MHz (20MHz) CH 7
WPA2
-62
dBm
Galaxy M13
WiFi 4
2e:f5:ac:44:9e:36 17m
2462MHz (20MHz) CH 11
WPA2
-65
dBm
Redmi Note 13 Pro+ 5G
WiFi 6
76:c0:81:f5:ed:de 48m
2467MHz (20MHz) CH 12
WPA2
-74
dBm


Access Points
2.4
5
6
GHz
PRO
IN RANGE 1
by Strength
iPhone
WiFi 6
3a:b4:b8:12:79:d6 0m
5745MHz (80MHz) CH 149(155)
WPA3
-28
dBm

Simple MATLAB Code clc;

clear;

close all;

```
% Distance from Wi-Fi source in metres
distance = [0 8 12 17 48];

% Measured RSSI values in dBm
RSSI = [-28 -58 -62 -65 -74];

% Plot graph
figure;

plot(distance, RSSI, '-o', 'LineWidth', 2,
'MarkerSize', 7);

grid on;

% Labels and title
xlabel('Distance from Wi-Fi Source (m)');
ylabel('Signal Strength (RSSI) (dBm)');
title('Wi-Fi Signal Strength vs Distance');

% Display RSSI values on graph
for i = 1:length(distance)
   text(distance(i), RSSI(i), ...
      sprintf(' %.0f dBm', RSSI(i)));
end
```

MATLAB

matlab.mathworks.com

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Prajwal

New Open Save Compare Go To Find Bookmark Refactor Code Issues Debugger Run Section Section Break Run and Advance Run to End Run Step Stop

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MATLAB Drive

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/MATLAB Drive/A27_Exp5_WMC.m

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Figure 1 ×

